

Appendix B

Protective Device Classification

The devices in switching equipment are referred to by numbers, with appropriate suffix letters when necessary, according to the functions they perform.

These numbers are based on a system adopted as standard for automatic switchgear by IEEE, and incorporated in American Standard C37.2-2008. This system is used in connection diagrams, in instruction books, and in specifications.

B.1 Device Function Numbers

See Table B.1.

B.2 Devices Performing More than One Function

If one device performs two relatively important functions in an equipment so that it is desirable to identify both of these functions, this may be done by using a double function number and name such as 50/51 instantaneous and time-overcurrent relay.

In many cases a single digital device may perform multiple functions. Such a device is known as a multifunction device. For multifunction devices, Standard C37.2-2008 recommends using device number 11 plus additional information that provides more detail as to what functions inside the device are used. For instance, the standard proposes a specific device number 11L-xxxx. In this proposal, “11” indicates multifunction, “L” indicates that in this example the device protects a line, and xxxx is the manufacturer’s model designation. Other examples could use “B” if the device protects a bus, or “T” if a transformer for instance. Additional functional information could be presented in one of three formats.

An “empty box” format showing analog and digital inputs and outputs. This format minimizes space requirements and also eliminates the need to update the device if the functionality changes. However the empty box provides no information as to what functions are actually used, and will usually also refer to other sources of information such as a logic drawing.

A “filled box” that diagrams individual functions and basic measurement and logic connections within the box. This option provides more information as to what functions are actually used, but takes up more space, and will need to be updated when the functionality changes.

A “List box” that identifies the functions within the box by function number, but includes no indication of measurement or logical connections. This option is a compromise between options (i) and (ii) in providing more information than option (i), but taking up less space than (ii).

Table B.1 Function numbers.

Device No.	Function	Device No.	Function	Device No.	Function
1	Master element	33	Position switch	65	Governor
2	Time-delay starting, or closing relay	34	Master sequence device.	66	Notching or jogging device
3	Checking or interlocking relay	35	Brush-operating, or slip-ring-short-circuiting device	67	Ac directional overcurrent relay
4	Master contactor	36	Polarity or polarizing voltage device	68	Blocking or “out-of-step”
5	Stopping device	37	Undercurrent or underpower relay	69	Permissive control device
6	Starting circuit breaker	38	Bearing protective device	70	Rheostat
7	Rate of change relay	39	Mechanical condition monitor	71	Liquid level switch
8	Control power disconnecting device	40	Field (over/under excitation) relay	72	DC circuit breaker
9	Reversing device	41	Field circuit breaker	73	Load-resistor contactor
10	Unit sequence switch	42	Running circuit breaker	74	Alarm relay
11	Multifunction device	43	Manual transfer or selector device	75	Position changing mechanism
12	Over-speed device	44	Unit-sequence starting relay	76	DC overcurrent relay.
13	Synchronous-speed device	45	Abnormal atmospheric condition monitor	77	Telemetry device
14	Under-speed device	46	Reverse-phase, or phase-balance, current relay	78	Phase angle measuring relay
15	Speed or frequency-matching device	47	Phase-sequence or phase-balance voltage relay	79	AC reclosing relay
16	Data communications device.	48	Incomplete sequence relay	80	Flow switch
17	Shunting or discharge switch	49	Machine, or transformer, thermal relay	81	Frequency relay
18	Accelerating or decelerating device	50	Instantaneous overcurrent relay	82	DC load-measuring reclosing relay
19	Starting-to-running transition contactor	51	AC inverse time overcurrent relay	83	Automatic selective control or transfer relay
20	Electrically operated valve	52	AC circuit breaker	84	Operating mechanism
21	Distance relay	53	Field excitation relay	85	Pilot communications carrier or pilot-wire relay
22	Equalizer circuit breaker	54	Turning gear engaging device	86	Lockout relay

Table B.1 (Continued)

Device No.	Function	Device No.	Function	Device No.	Function
23	Temperature control device	55	Power factor relay	87	Differential protective relay
24	Volts per Hertz relay	56	Field application relay	88	Auxiliary motor or motor generator
25	Synchronizing or synchronism-check device	57	Short-circuiting or grounding device	89	Line switch
26	Apparatus thermal device	58	Rectification failure relay	90	Regulating device
27	Undervoltage relay	59	Overvoltage relay	91	Voltage directional relay
28	Flame detector	60	Voltage or current balance relay	92	Voltage and power directional relay
29	Isolating contactor	61	Density switch or sensor	93	Field changing contactor
30	Annunciator relay	62	Time-delay stopping or opening relay	94	Tripping or trip-free relay
31	Separate excitation device	63	Pressure switch	95–99.	Used only for specific applications
32	Directional power relay	64	Ground detector relay		

Notwithstanding the options for multifunction devices presented by Standard C37.2-2008, some users choose to use the number of the primary function in a multifunction device and rely on other sources (drawings and settings information) to identify what and how other functions are used. For example, a transmission line current differential relay might be designated “87L” even though it includes numerous other functions such as distance, overcurrent, voltage, automatic reclosing, and logic functions.

B.2.1 Suffix Numbers

If two or more devices with the same function number and suffix letter (if used) are present in the same equipment, they may be distinguished by numbered suffixes as for example, 52X-1, 52X-2 and 52X-3, when necessary.

B.2.2 Suffix Letters

Suffix letters are used with device function numbers for various purposes. In order to prevent possible conflict each suffix letter should have only one meaning in an individual equipment. All other words should use the abbreviations as contained in the latest revision of the standard ASME Y14.38 “Abbreviations and Acronyms for Use on Drawings and Related Documents,” or should use some other distinctive abbreviation, or be written out in full each time they are used. The meaning of each single suffix letter, or combination of letters, should be clearly designated in the legend on the drawings or publications applying to the equipment.

Lowercase (small) suffix letters are used in practically all instances on electrical diagrams for the auxiliary, position, and limit switches. Capital letters are generally used for all other suffix letters.

The letters should generally form part of the device function designation and are usually written directly after the device function number, as, for example, 52CS, 71W, or 49D. When it is necessary to use two types of suffix letters in connection with one function number, it is often desirable for clarity to separate them by a slanted line or dash, as, for example, 2-D/CS or 20D-CS.

The suffix letters that denote parts of the main device, and those that cannot or need not form part of the device function destination, are generally written directly below the device function number on drawings, as for example,

$$\frac{52}{CC} \quad \text{or} \quad \frac{43}{A}$$

All auxiliary contacts, position, and limit switches for devices and equipment as circuit breakers, contactors, valves and rheostats, and contacts of relays are designated as follows:

a	Contact that is open when the main device is in that standard reference position, commonly referred to as the non-operated or de-energized position, that closes when the device assumes the opposite position
b	Contact that is closed when the main device is in the standard reference position, commonly referred to as the non-operated or de-energized position, and that opens when the device assumes the opposite position
aa	Contact that is open when the operating mechanism of the main device is in the non-operated position and that closes when the operating mechanism assumes the opposite position
bb	Contact that is closed when the operating mechanism of the main device is in the non-operated position and that opens when the operating mechanism assumes the opposite position

B.2.3 Representation of Device Contacts on Electrical Diagrams

On electrical diagrams the “b” contacts of all devices, including those of relays and those with suffix letters or percentage figures, should be shown as closed contacts, and all “a” contacts should be shown as open contacts.

For those devices that have no de-energized or non-operated position, such as manually operated transfer or control switches (including those of the spring-return type) or auxiliary position indicating contacts on the housings or enclosures of a removable circuit breaker unit, the preferred method of representing these contacts is as an “a” switch. Each contact should, however, be identified on the elementary diagram as to when it closes.

In the case of latched-in or hand-reset locking-out relays, which operate from protective devices to perform the shutdown of an equipment and to hold it out of service, the contacts should preferably be shown in the normal non-locking-out position. In general, any devices, such as electrically operated latched-in relays, which have not been specifically covered in the aforementioned paragraphs should have their contacts shown in the position most suitable for the ready understanding of the operation of the devices in the equipment, and sufficient description should be present, as necessary, on the elementary diagram to indicate the contact operation.